

Minsoo Cheong

Integrated M.S./Ph.D. Student in Computer Science and Engineering

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RESEARCH INTERESTS

Efficient machine learning systems; language model inference and serving; KV-cache optimization; low-bit quantization; diffusion language models.

EDUCATION

Seoul National University · Seoul, Republic of Korea

Sep 2025 - Present

Integrated M.S./Ph.D. in Computer Science and Engineering

Advisor: Prof. Sungjoo Yoo | Computing and Memory Architecture Laboratory (CMALab)

Seoul National University · Seoul, Republic of Korea

Mar 2019 - Aug 2025

B.S. in Computer Science and Engineering, *Cum Laude*

PUBLICATIONS

[[NeurIPS 2026](#)] EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models [[Paper](#)] [[Code](#)]

Minsoo Cheong, Donghyun Son, Woosang Lim, and Sungjoo Yoo

[[Preprint](#)] Tailoring the Quantization Space for 1-Bit KV Cache Compression [[Project Page](#)] [[Paper](#)] [[Code](#)]

Minsoo Cheong*, Donghyun Son*, and Sungjoo Yoo | * *Equal contribution.*

[[Preprint](#)] KV-Rescue: Recovering Reasoning Language Model KV Eviction Loss via Stepwise Interleaving [[Paper](#)]

Minsoo Cheong, Woosang Lim, Vincent-Daniel Yun, and Sungjoo Yoo

[[Preprint](#)] Output-Aware Rotation for INT2 KV-Cache Quantization [[Paper](#)] [[Code](#)]

Vincent-Daniel Yun, Woosang Lim, Minsoo Cheong, Sunwoo Lee, Murali Annavaram, Sai Praneeth Karimireddy, and Sungjoo Yoo

PRESENTATIONS AND ACADEMIC SERVICE

EntropyCache. Poster presentations at NeurIPS 2026 and the ICML 2026 AdaptFM Workshop; talk at the Samsung MX Research Project Exchange Seminar, Jul 2026.

Reviewer, NeurIPS 2026 Workshop on On-Device Intelligence: Foundation Models under Real-World Constraints (ODI).

OPEN SOURCE CONTRIBUTIONS

llama.cpp | Collaborator

Mar 2024 - Present

Contributions to stochastic speculative decoding, the retrieval example, and EXAONE model support in the widely used C/C++ LLM inference runtime. [[Contributions](#)]

Spring Framework, Armeria, gRPC & Protocol Buffers | Contributor

Contributions spanning Java/Kotlin frameworks, RPC infrastructure, and serialization tooling. [[Contributions](#)]

krotoDC

Open-source project

A protoc plugin that generates Kotlin data classes and coroutine-friendly gRPC services and stubs, making Protocol Buffers and gRPC more idiomatic in Kotlin; distributed through Maven Central with 5,000+ monthly downloads. [[GitHub](#)]

INDUSTRY EXPERIENCE

NAVER Corporation · Research Engineer Intern, LLM Solution Team

Jun 2024 - Aug 2024

Built a KubeFlow pipeline for automated LLM inference benchmarking across hardware, batching, sequence length, quantization, and speculative decoding; implemented llama.cpp support for an internal language model.

Mathpresso · Backend Software Engineer, Content Platform Team

Jun 2022 - Jun 2023

Built Kotlin/Go content-production and distribution systems and gRPC APIs; contributed to three release cycles of a digital school-exam product that achieved a 60-fold increase in sales.

Bucketplace · Backend Software Engineer, Commerce Service Team

May 2021 - Jun 2022

Developed commerce backend services using Kotlin and Spring Boot, including bulk product management and microservice data aggregation.

TECHNICAL SKILLS

Machine Learning & Systems: PyTorch, CUDA, Hugging Face Transformers, LLM inference, KV caching, model efficiency

Languages & Infrastructure: Python, C++, Kotlin, Java; Git, Protocol Buffers, gRPC, Spring Boot, Project Reactor, Kotlin Coroutines